

Dapaquis 10 mg Film-Coated Tablets

Dapagliflozin 10 mg

COMPOSITION

Each film coated tablet contains dapagliflozin propanediol equivalent to dapagliflozin 10 mg

DOSAGE FORM

Film-coated tablet

Pink, biconvex, diamond-shaped, film -coated tablets with “C389” engraved on one side and plain on the other side.

PHARMACOLOGY

Pharmacodynamics

Pharmacotherapeutic group: Drugs used in diabetes, sodium-glucose co-transporter 2 (SGLT2) inhibitors, ATC code: A10BK01

Mechanism of action

Dapagliflozin is a highly potent (K_i : 0.55 nM), selective and reversible inhibitor of SGLT2.

Inhibition of SGLT2 by dapagliflozin reduces reabsorption of glucose from the glomerular filtrate in the proximal renal tubule with a concomitant reduction in sodium reabsorption leading to urinary excretion of glucose and osmotic diuresis. Dapagliflozin therefore increases the delivery of sodium to the distal tubule which is believed to increase tubuloglomerular feedback and reduce intraglomerular pressure. This combined with osmotic diuresis leads to a reduction in volume overload, reduced blood pressure, and lower preload and afterload, which may have beneficial effects on cardiac remodelling. Other effects include an increase in haematocrit and reduction in body weight. The cardiac benefits of dapagliflozin are not solely dependent on the blood glucose-lowering effect and not limited to patients with diabetes..

Dapagliflozin improves both fasting and post-prandial plasma glucose levels by reducing renal glucose reabsorption leading to urinary glucose excretion. This glucose excretion (glucuretic effect) is observed after the first dose, is continuous over the 24-hour dosing interval and is sustained for the duration of treatment. The amount of glucose removed by the kidney through this mechanism is dependent upon the blood glucose concentration and GFR. Thus, in patients with normal blood glucose, dapagliflozin has a low propensity to cause hypoglycaemia. Dapagliflozin does not impair normal endogenous glucose production in response to hypoglycaemia. Dapagliflozin acts independently of insulin secretion and insulin

action. Improvement in homeostasis model assessment for beta cell function (HOMA beta-cell).

The SGLT2 is selectively expressed in the kidney. Dapagliflozin does not inhibit other glucose transporters important for glucose transport into peripheral tissues and is > 1,400 times more selective for SGLT2 versus SGLT1, the major transporter in the gut responsible for glucose absorption.

Pharmacodynamic effects

There is an increase in the amount of glucose excreted in the urine following the administration of dapagliflozin.

This urinary glucose excretion with dapagliflozin also results in osmotic diuresis and increases in urinary volume. The increase in urinary volume was associated with a small and transient increase in urinary sodium excretion that was not associated with changes in serum sodium concentrations.

Urinary uric acid excretion was also increased transiently (for 3-7 days) and accompanied by a sustained reduction in serum uric acid concentration.

Pharmacokinetics

Absorption

Dapagliflozin was rapidly and well absorbed after oral administration. Maximum dapagliflozin plasma concentrations ($C_{\max}$) were usually attained within 2 hours after administration in the fasted state. Geometric mean steady-state dapagliflozin $C_{\max}$ and AUC_{τ} values following once daily 10 mg doses of dapagliflozin were 158 ng/mL and 628 ng h/mL, respectively. The absolute oral bioavailability of dapagliflozin following the administration of a 10-mg dose is 78%. Administration with a high-fat meal decreased dapagliflozin $C_{\max}$ by up to 50% and prolonged $T_{\max}$ by approximately 1 hour, but did not alter AUC as compared with the fasted state. These changes are not considered to be clinically meaningful. Hence, dapagliflozin can be administered with or without food.

Distribution

Dapagliflozin is approximately 91% protein bound. Protein binding was not altered in various disease states (e.g. renal or hepatic impairment). The mean steady-state volume of distribution of dapagliflozin was 118 litres.

Biotransformation

Dapagliflozin is extensively metabolised, primarily to yield dapagliflozin 3-O-glucuronide, which is an inactive metabolite. Dapagliflozin 3-O-glucuronide or other metabolites do not contribute to the glucose-lowering effects. The formation of dapagliflozin 3-O-glucuronide is mediated by UGT1A9, an enzyme present in the liver and kidney, and CYP-mediated metabolism was a minor clearance pathway in humans.

Elimination

The mean plasma terminal half-life ($t_{1/2}$) for dapagliflozin was 12.9 hours following a single oral dose of dapagliflozin 10 mg. The mean total systemic clearance of dapagliflozin administered intravenously was 207 mL/min. Dapagliflozin and related metabolites are primarily eliminated via urinary excretion with less than 2% as unchanged dapagliflozin. In faeces, approximately 15% of the dose was excreted as parent drug.

Linearity

Dapagliflozin exposure increased proportional to the increment in dapagliflozin dose over the range of 0.1 to 500 mg and its pharmacokinetics did not change with time upon repeated daily dosing for up to 24 weeks.

Special populations

Renal impairment

At steady-state (20 mg once-daily dapagliflozin for 7 days), there is an increase in mean systemic exposures of dapagliflozin in patients with type 2 diabetes and renal impairment. The steady-state 24-hour urinary glucose excretion was highly dependent on renal function. The impact of hemodialysis on dapagliflozin exposure is not known.

Hepatic impairment

There is no clinically meaningful impact on pharmacokinetics of dapagliflozin in patients with mild or moderate hepatic impairment (Child-Pugh classes A and B). There is an increase in mean C_{max} and AUC of dapagliflozin in patients with severe hepatic impairment (Child-Pugh class C).

Elderly (≥ 65 years)

There is no clinically meaningful increase in exposure based on age alone in elderly up to 70 years old. However, an increased exposure due to age-related decrease in renal function can be expected. There are insufficient data to draw conclusions regarding exposure in patients > 70 years old.

Paediatric population

Pharmacokinetics in the paediatric population have not been studied.

Gender

The mean dapagliflozin AUC_{ss} in females was estimated to be about 22% higher than in males.

Race

There were no clinically relevant differences in systemic exposures between White, Black or Asian races.

Body weight

Dapagliflozin exposure was found to decrease with increased weight. Consequently, low-weight patients may have somewhat increased exposure and patients with high weight somewhat decreased exposure. However, the differences in exposure were not considered clinically meaningful.

INDICATIONS

Type 2 diabetes mellitus

Dapagliflozin is indicated in adults aged 18 years and older with type 2 diabetes mellitus as:

Monotherapy

When diet and exercise alone do not provide adequate glycaemic control in patients for whom use of metformin is considered inappropriate due to intolerance.

Add-on combination therapy

In combination with other glucose-lowering medicinal products including insulin, when these, together with diet and exercise, do not provide adequate glycaemic control (see *Warnings and Precautions*, *Drug Interactions* and *Pharmacodynamics* for available data on different combinations).

Initial Combination

Dapagliflozin is indicated for use as initial combination therapy with metformin, as an adjunct to diet and exercise, to improve glycemic control in patients with type 2 diabetes mellitus when diet and exercise have failed to provide adequate glycemic control and there are poor prospects for response to metformin monotherapy (for example, high initial HbA1c levels).

To reduce the risk of hospitalization for heart failure in adults with type 2 diabetes mellitus and established cardiovascular disease (CVD) or multiple cardiovascular (CV) risk factors.

Heart failure

Dapagliflozin is indicated in adults for the treatment of symptomatic chronic heart failure.

Chronic kidney disease

Dapagliflozin is indicated in adults for the treatment of chronic kidney disease.

DOSAGE AND METHOD OF ADMINISTRATION

Posology

Type 2 diabetes mellitus

Monotherapy and add-on combination therapy

The recommended dose is 10 mg dapagliflozin once daily.

When dapagliflozin is used in combination with insulin or an insulin secretagogue, such as a sulphonylurea, a lower dose of insulin or insulin secretagogue may be considered to reduce the risk of hypoglycaemia (see *Drug Interactions* and *Undesirable Effects*).

Initial Combination Therapy

The recommended starting doses of dapagliflozin and metformin when used as initial combination therapy are 10 mg dapagliflozin plus 500 mg metformin once daily. Patients with inadequate glycemic control on this dose should further have their metformin dose increased according to approved local label guidelines.

Heart failure

The recommended dose is 10mg dapagliflozin once daily.

Chronic kidney disease

The recommended dose is 10mg dapagliflozin once daily.

Special populations

Renal impairment

No dosage adjustment is required based on renal function.

Due to limited experience, it is not recommended to initiate treatment with dapagliflozin in patients with GFR <25 mL/min.

In patients with diabetes mellitus, the glucose lowering efficacy of dapagliflozin is reduced when the glomerular filtration rate is <45 mL/min, and is likely absent in patients with severe renal impairment. Therefore, if GFR falls below 45 mL/min, additional glucose lowering treatment should be considered in patients with diabetes mellitus if further glycaemic control is needed.

Hepatic impairment

No dosage adjustment is necessary for patients with mild or moderate hepatic impairment. In patients with severe hepatic impairment, a starting dose of 5 mg is recommended. If well tolerated, the dose may be increased to 10 mg (see *Warnings and Precautions* and *Pharmacokinetics*).

Elderly (≥ 65 years)

No dosage adjustment is recommended based on age.

Paediatric population

The safety and efficacy of dapagliflozin in children aged 0 to < 18 years have not yet been established. No data are available.

Dapaquis is available at the strengths of 10 mg only and may not be able to deliver all the dosing recommendations mentioned above. In such cases, other approved strengths should be used.

Method of administration

Dapagliflozin can be taken orally once daily at any time of day with or without food. Tablets are to be swallowed whole.

CONTRAINDICATIONS

Hypersensitivity to the active substance or to any of the excipients.

WARNING AND PRECAUTIONS

Renal impairment

Due to limited experience, it is not recommended to initiate treatment with dapagliflozin in patients with GFR <25 mL/min.

The glucose lowering efficacy of dapagliflozin is dependent on renal function, and is reduced in patients with GFR <45 mL/min and is likely absent in patients with severe renal impairment.

Hepatic impairment

Dapagliflozin exposure is increased in patients with severe hepatic impairment (see *Dosage and Method of Administration* and *Pharmacokinetics*).

Use in patients at risk for volume depletion and/or hypotension

Due to its mechanism of action, dapagliflozin increases diuresis which may lead to the modest decrease in blood pressure (see *Pharmacodynamics*). It may be more pronounced in patients with very high blood glucose concentrations.

Caution should be exercised in patients for whom a dapagliflozin-induced drop in blood pressure could pose a risk, such as patients on anti-hypertensive therapy with a history of hypotension or elderly patients.

In case of intercurrent conditions that may lead to volume depletion (e.g. gastrointestinal illness), careful monitoring of volume status (e.g. physical examination, blood pressure measurements, laboratory tests including haematocrit and electrolytes) is recommended. Temporary interruption of treatment with dapagliflozin is recommended for patients who develop volume depletion until the depletion is corrected (see *Undesirable Effects*).

Diabetic ketoacidosis

Sodium-glucose co-transporter 2 (SGLT2) inhibitors should be used with caution in patients with increased risk of DKA. Patients who may be at higher risk of DKA include patients with a low beta-cell function reserve (e.g. type 2 diabetes patients with low C-peptide or latent autoimmune diabetes in adults (LADA) or patients with a history of pancreatitis), patients with conditions that lead to restricted food intake or severe dehydration, patients for whom insulin doses are reduced and patients with increased insulin requirements due to acute medical illness, surgery or alcohol abuse.

The risk of diabetic ketoacidosis must be considered in the event of non-specific symptoms such as nausea, vomiting, anorexia, abdominal pain, excessive thirst, difficulty breathing, confusion, unusual fatigue or sleepiness. Patients should be assessed for ketoacidosis immediately if these symptoms occur, regardless of blood glucose level.

Before initiating dapagliflozin, factors in the patient history that may predispose to ketoacidosis should be considered.

Treatment should be interrupted in patients who are hospitalised for major surgical procedures or acute serious medical illnesses. In both cases, treatment with dapagliflozin may be restarted once the patient's condition has stabilised.

Type 2 diabetes mellitus

In patients where DKA is suspected or diagnosed, treatment with dapagliflozin should be discontinued immediately.

Restarting SGLT2 inhibitor treatment in patients experiencing a DKA while on SGLT2 inhibitor treatment is not recommended, unless another clear precipitating factor is identified and resolved.

Type 1 diabetes mellitus

Dapagliflozin should not be used for treatment of patients with type 1 diabetes.

Necrotising fasciitis of the perineum (Fournier's gangrene)

This is a rare but serious and potentially life-threatening event that requires urgent surgical intervention and antibiotic treatment.

Patients should be advised to seek medical attention if they experience a combination of symptoms of pain, tenderness, erythema, or swelling in the genital or perineal area, with fever

or malaise. Be aware that either uro-genital infection or perineal abscess may precede necrotizing fasciitis. If Fournier's gangrene is suspected, dapagliflozin should be discontinued and prompt treatment (including antibiotics and surgical debridement) should be instituted.

Urinary tract infections

Urinary glucose excretion may be associated with an increased risk of urinary tract infection; therefore, temporary interruption of dapagliflozin should be considered when treating pyelonephritis or urosepsis.

Elderly (≥ 65 years)

Elderly patients may be at a greater risk for volume depletion and are more likely to be treated with diuretics.

Elderly patients are more likely to have impaired renal function, and/or to be treated with anti-hypertensive medicinal products that may cause changes in renal function such as angiotensin-converting enzyme inhibitors (ACE-I) and angiotensin II type 1 receptor blockers (ARB). The same recommendations for renal function apply to elderly patients as to all patients (see *Dosage and Method of Administration, Warnings and Precautions, Undesirable Effects* and *Pharmacodynamics*).

Cardiac failure

Experience with dapagliflozin in NYHA class IV is limited.

Chronic kidney disease

There is no experience with dapagliflozin for the treatment of chronic kidney disease in patients without diabetes who do not have albuminuria. Patients with albuminuria may benefit more from treatment with dapagliflozin.

Lower limb amputations

An increase in cases of lower limb amputation (primarily of the toe) has been observed in long-term, in type 2 diabetes mellitus with another SGLT2 inhibitor. It is unknown whether this constitutes a class effect. It is important to counsel patients with diabetes on routine preventative foot care.

Urine laboratory assessments

Due to its mechanism of action, patients taking dapagliflozin will test positive for glucose in their urine.

Lactose

The tablets contain lactose. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency, or glucose-galactose malabsorption should not take this medicinal product.

DRUG INTERACTIONS

Pharmacodynamic interactions

Diuretics

Dapagliflozin may add to the diuretic effect of thiazide and loop diuretics and may increase the risk of dehydration and hypotension (see *Warnings and Precautions*).

Insulin and insulin secretagogues

Insulin and insulin secretagogues, such as sulphonylureas, cause hypoglycaemia. Therefore, a lower dose of insulin or an insulin secretagogue may be required to reduce the risk of hypoglycaemia when used in combination with dapagliflozin (see *Dosage and Method of Administration* and *Undesirable Effects*).

Pharmacokinetic interactions

The metabolism of dapagliflozin is primarily via glucuronide conjugation mediated by UDP glucuronosyltransferase 1A9 (UGT1A9).

Dapagliflozin neither inhibited cytochrome P450 (CYP) 1A2, CYP2A6, CYP2B6, CYP2C8, CYP2C9, CYP2C19, CYP2D6, CYP3A4, nor induced CYP1A2, CYP2B6 or CYP3A4. Therefore, dapagliflozin is not expected to alter the metabolic clearance of coadministered medicinal products that are metabolised by these enzymes.

Effect of other medicinal products on dapagliflozin

The pharmacokinetics of dapagliflozin are not altered by metformin, pioglitazone, sitagliptin, glimepiride, voglibose, hydrochlorothiazide, bumetanide, valsartan, or simvastatin.

No dose adjustment is recommended following coadministration of dapagliflozin with rifampicin (an inducer of various active transporters and drug-metabolising enzymes).

No dose adjustment is recommended following coadministration of dapagliflozin with mefenamic acid (an inhibitor of UGT1A9).

Effect of dapagliflozin on other medicinal products

Dapagliflozin did not alter the pharmacokinetics of metformin, pioglitazone, sitagliptin, glimepiride, hydrochlorothiazide, bumetanide, valsartan, digoxin (a P-gp substrate) or warfarin (S-warfarin, a CYP2C9 substrate), or the anticoagulatory effects of warfarin as measured by INR. There is no significant impact on simvastatin exposure during combination of a single dose of dapagliflozin 20 mg and simvastatin (a CYP3A4 substrate).

Other interactions

The effects of smoking, diet, herbal products and alcohol use on the pharmacokinetics of dapagliflozin have not been studied.

Interference with 1,5-anhydroglucitol (1,5-AG) assay

Monitoring glycemic control with 1,5-AG assay is not recommended as measurements of 1,5-AG are unreliable in assessing glycemic control in patients taking SGLT2 inhibitors. Use of alternative methods to monitor glycemic control is advised.

Paediatric population

Interaction studies have only been performed in adults.

FERTILITY, PREGNANCY AND LACTATION

Pregnancy

The use of dapagliflozin is not recommended during the second and third trimesters of pregnancy.

When pregnancy is detected, treatment with dapagliflozin should be discontinued.

Breast-feeding

It is unknown whether dapagliflozin and/or its metabolites are excreted in human milk. A risk to the newborns/infants cannot be excluded. Dapagliflozin should not be used while breast-feeding.

Fertility

The effect of dapagliflozin on fertility in humans has not been studied.

UNDESIRABLE EFFECTS

Adverse reactions listed below are classified according to frequency and system organ class (SOC). Frequency categories are defined according to the following convention: very common, common, uncommon, rare), very rare, and not known.

Table 1. Adverse reactions

System organ class	Very common	Common	Uncommon	Rare	Very rare
<i>Infections and infestations</i>		Vulvovaginitis, balanitis and related genital infections Urinary tract infection	Fungal infection		Necrotising fasciitis of the perineum (Fournier's Gangrene)
<i>Metabolism and nutrition disorders</i>	Hypoglycaemia (when used with SU or		Volume depletion Thirst	Diabetic ketoacidosis (when used in type 2 diabetes	

	insulin)			mellitus	
<i>Nervous system disorders</i>		Dizziness			
<i>Gastrointestinal disorders</i>			Constipation Dry mouth		
<i>Skin and subcutaneous tissue disorders</i>		Rash			
<i>Musculoskeletal and connective tissue disorders</i>		Back pain			
<i>Renal and urinary disorders</i>		Dysuria Polyuria	Nocturia		
<i>Reproductive system and breast disorders</i>			Vulvovaginal pruritus Pruritus genital		
<i>Investigations</i>		Haematocrit increased, Creatinine renal clearance decreased during initial treatment, Dyslipidaemia	Blood creatinine increased during initial treatment, Blood urea increased, Weight decreased		

Reporting of suspected adverse reactions

Reporting suspected adverse reaction after authorisation of the medicinal product is important. It allows continued monitoring of the benefit / risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the local reporting system.

OVERDOSE

Dapagliflozin did not show any toxicity at single oral doses up to 500 mg (50 times the maximum recommended human dose). In the event of an overdose, appropriate supportive

treatment should be initiated as dictated by the patient's clinical status. The removal of dapagliflozin by haemodialysis has not been studied.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Dapaquis has no or negligible influence on the ability to drive and use machines. Patients should be alerted to the risk of hypoglycaemia when dapagliflozin is used in combination with a sulphonylurea or insulin.

INCOMPATIBLITY

Not applicable

STORAGE AND HANDLING INSTRUCTIONS

Store below 30°C.

SHELF LIFE

24 months

PACKAGING INFORMATION

One carton containing one HDPE bottle of 30 tablets.

PRODUCT REGISTRATION HOLDER

Cipla Malaysia Sdn Bhd
Suite 1101, Amcorp Tower,
Amcorp Trade Centre,
18 Persiaran Barat,
46050 Petaling Jaya,
Selangor, Malaysia

MANUFACTURED BY

Cipla Ltd.
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DATE OF REVISION

December 2023

Date: 14th March 2022

To,
National Pharmaceutical Regulatory Agency,
Lot 36, Jalan Universiti, 46200,
Petaling Jaya, Selangor,
Malaysia

Declaration Letter on Font Size and Font Type for Package Insert

We would like to inform that currently for Pack insert we are submitting PDF version file for evaluation in dossier.

During commercialisation we would be using font size 7 pt and font type Arial or Helvetica for commercial layout of Pack insert. This font type gives clear printing and readability for leaflet copy matter.

Please consider above clarification and approve the same.

Thanking you,

For Cipla Ltd.,



Date: 14.03.2022 18:31:00 +05:30

Sheetal More

Senior Manager Regulatory Affairs